

Step-by-Step Guide to Use Label Studio for Annotation

What is Label Studio?

Label Studio is an open-source data annotation platform that supports:

- Classification
- Object Detection (Bounding Boxes)
- Segmentation
- Keypoint Annotation
- Text Annotation

In this tutorial, we will use Label Studio to annotate x-ray images using bounding boxes.

1. Install Label Studio

Windows

Step 1: Install Python

Download Python (3.11 or 3.12 recommended):

<https://www.python.org/downloads/>

During installation, check:

✓ Add Python to PATH

Step 2: Open Command Prompt

```
pip install label-studio
```

Step 3: Start Label Studio

```
label-studio start
```

macOS

Install Homebrew (if needed)

<https://brew.sh>

Create a Virtual Environment

```
mkdir labelstudio  
cd labelstudio
```

```
python3 -m venv venv  
source venv/bin/activate
```

Install Label Studio

```
pip install label-studio
```

Start Label Studio

```
label-studio start
```

Linux (Ubuntu)

Install Python

```
sudo apt update  
sudo apt install python3 python3-pip python3-venv
```

Create Environment

```
mkdir labelstudio  
cd labelstudio
```

```
python3 -m venv venv  
source venv/bin/activate
```

Install Label Studio

```
pip install label-studio
```

Start Label Studio

```
label-studio start
```

2. Open Label Studio

After starting Label Studio, open:

<http://localhost:8080>

Create an account if prompted.

3. Create a New Project

1. Click **Create Project**

2. Enter a project name

Example:

Knee Annotation

3. Click **Create**
-

4. Configure the Labeling Interface

Go to:

Settings → Labeling Interface

Paste:

```
<View>
  <Image name="image" value="$image"/>

  <RectangleLabels name="label" toName="image">
    <Label value="ROI"/>
  </RectangleLabels>
</View>
```

Click **Save**.

5. Upload x-ray Images

Upload:

1. Go to Tasks
 2. Click Import
 3. Select images
 4. Upload
-

6. Start Annotating

Open an image.

Draw Bounding Box

1. Select Rectangle Tool
2. Click and drag

3. Draw a box around the ACL tear
4. Select label “ROI”

Save Annotation

Click:

Submit

or

Ctrl + Enter

8. Keyboard Shortcuts

Action	Shortcut
Submit Annotation	Ctrl + Enter
Zoom In	Mouse Wheel Up
Zoom Out	Mouse Wheel Down
Pan Image	Hold Space
Delete Region	Delete

9. Quality Control

Before exporting:

- Verify every image is annotated
- Check box placement
- Ensure label consistency
- Review difficult cases

Common mistakes:

- Missing annotations
 - Inaccurate bounding boxes
 - Incorrect labels
 - Duplicate boxes
-

10. Export Annotations

After annotation:

1. Go to Export
2. Select format

Common formats:

- COCO JSON
- YOLO
- Pascal VOC XML

Recommended:

- YOLO → Ultralytics YOLO
 - COCO → Detectron2
 - Pascal VOC → Traditional pipelines
3. Click Download
-

Helpful Links

Label Studio Official Website

<https://labelstud.io/>

Documentation

<https://labelstud.io/guide/>

GitHub

<https://github.com/HumanSignal/label-studio>

Python Downloads

<https://www.python.org/downloads/>

Homebrew

<https://brew.sh>